The sharp one-dimensional convex sub-Gaussian comparison constant

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Abstract

Let X be an integrable real random variable with mean zero and two-sided sub-Gaussian tail $\mathbb{P}(|X| > t) \leq 2e^{-t^2/2}$ for all $t \geq 0$. We determine the smallest constant $c_\star$ such that X is dominated in convex order by $c_\star G$, where G is standard normal. Equivalently, $c_\star^2$ is the sharp one-dimensional convex sub-Gaussian comparison constant appearing in the *Optimization Constants in Mathematics* repository [DITc26]. We show that $c_\star$ is given by an explicit system of one-dimensional equations and is attained by an extremal distribution that saturates the tail constraint. Numerically, $c_\star \approx 2.30952$ (so $c_\star^2 \approx 5.33386$). We also determine the analogous sharp constant under a two-sided sub-exponential tail bound, with convex domination by a scaled Laplace law. Finally, we record two higher-dimensional consequences: a sequential tensorization principle for multivariate convex domination, and a dimension-free Gaussian comparator for the cone generated by convex ridge functions (the linear convex order).

1 Introduction

A recent theorem of van Handel [vH25] shows that if X is a random vector in $\mathbb{R}^d$ such that $\mathbb{P}(|\langle v, X\rangle| > t) \leq 2e^{-t^2/2}$ for all $\|v\|_2 = 1$ and $t \geq 0$, then X is dominated in convex order by a universal constant times a standard Gaussian vector. The optimal value of this universal constant is not known, even in dimension 1.

This note resolves the one-dimensional case. We compute the sharp constant and exhibit an extremal distribution. The argument is elementary and rests on two classical facts: (i) one-dimensional convex order is equivalent to comparison of the stop-loss transforms $u \mapsto \mathbb{E}[(X-u)_+]$ [SS07, Ch. 3]; (ii) under a two-sided tail constraint, the stop-loss transform is maximized by a distribution that saturates the constraint and has a single flat region.

Setup and the constant

Throughout, $G \sim \mathcal{N}(0,1)$, $\varphi(x) := (2\pi)^{-1/2}e^{-x^2/2}$ is the standard normal density, and $\overline{\Phi}(x) := \int_x^\infty \varphi(t)\,\mathrm{d}t$ is the Gaussian tail. We call X 1-*sub-Gaussian in the tail sense* if

$$\mathbb{E}[X] = 0 \qquad \text{and} \qquad \mathbb{P}(|X| > t) \leq s_G(t) := \min\{1, 2e^{-t^2/2}\} \qquad \text{for all } t \geq 0. \tag{1}$$

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Define the one-dimensional comparison constant

$$c_\star := \inf\Big\{c > 0 : \text{ every } X \text{ satisfying (1) obeys } X \preceq_{cx} cG\Big\}, \tag{2}$$

where $X \preceq_{cx} Y$ denotes convex domination: $\mathbb{E}[f(X)] \le \mathbb{E}[f(Y)]$ for every convex $f : \mathbb{R} \to \mathbb{R}$ for which both expectations are finite.

Let $t_0 := \sqrt{2\log 2}$ denote the point at which s_G first descends from 1, and hence define

$$A := \int_0^\infty s_G(t)\,\mathrm{d}t = t_0 + 2\int_{t_0}^\infty e^{-t^2/2}\,\mathrm{d}t, \qquad B := \frac{A}{2}. \tag{3}$$

For $x \ge t_0$, set

$$H(x) := 2x\,e^{-x^2/2} + 2\int_x^\infty e^{-t^2/2}\,\mathrm{d}t. \tag{4}$$

Since H is strictly decreasing with $H(t_0) = A$ and $\lim_{x\to\infty} H(x) = 0$, the intermediate value theorem supplies a unique $a > t_0$ such that $H(a) = B$. Define then

$$p_0 := 2e^{-a^2/2} \in (0,1), \tag{5}$$

let z be the unique solution of $\overline{\Phi}(z) = p_0$, and set

$$c_0 := \frac{B}{\varphi(z)}. \tag{6}$$

We now state the main result of this note.

Theorem 1 (Sharp one-dimensional convex sub-Gaussian comparison)**.** *The sharp constant in* (2) *satisfies $c_\star = c_0$, where c_0 is defined by* (3)–(6)*. In particular:*

1. *For every random variable X satisfying* (1) *and every convex $f : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{E}[f(X)] \le \mathbb{E}[f(c_0 G)] \tag{7}$$

   whenever the right-hand side is finite.

2. *For every $c < c_0$, there exist a random variable $X^\star$ satisfying* (1) *and a convex function f such that $\mathbb{E}[f(X^\star)] > \mathbb{E}[f(cG)]$. One may take $f(x) = (x - cz)_+$, where z is the parameter from* (6)*.*

Consequently, the one-dimensional value of the constant C_{48} in [DITc26] is $C_{48}^{(1)} = c_0^2$.

Remark 2 (Numerical value)**.** *A direct high-precision evaluation of* (3)–(6) *gives*

$$a \approx 1.80334, \qquad p_0 \approx 0.39342, \qquad z \approx 0.27041, \qquad c_0 \approx 2.30952, \qquad c_0^2 \approx 5.33386.$$

No numerical computation is used in the derivation of the exact characterization $c_\star = c_0$.

Remark 3 (Other notions of sub-Gaussianity)**.** *Note that a sub-Gaussian bound on the moment-generating function of X of the form*

$$\mathbb{E}[e^{\lambda X}] \le e^{\lambda^2/2} \qquad \text{for all } \lambda \in \mathbb{R}$$

immediately implies the tail constraint with which we work. Whether sharper convex orderings can be found under this nominally stronger assumption is an interesting question, and seems likely to admit different extrema.

Section 5 records the analogous sharp constant under a two-sided sub-exponential tail bound, with convex domination by a scaled Laplace law. The case of general $d \ge 2$ remains open.

2 Convex order and stop-loss transforms

We recall the characterization of one-dimensional convex order in terms of stop-loss transforms. For $u \in \mathbb{R}$ define the hinge function $(x-u)_+ := \max\{x-u, 0\}$.

Proposition 4 (Stop-loss characterization of convex domination)**.** *Let X, Y be integrable real random variables. Then $X \preceq_{cx} Y$ if and only if*

$$\mathbb{E}[X] = \mathbb{E}[Y] \qquad \text{and} \qquad \mathbb{E}[(X-u)_+] \le \mathbb{E}[(Y-u)_+] \quad \text{for all } u \in \mathbb{R}. \tag{8}$$

Proof. This is standard; see, e.g., [SS07, Thm. 3.A.1]. For completeness, we recall the short direction needed below. Assume (8). Any convex $f : \mathbb{R} \to \mathbb{R}$ admits the representation

$$f(x) = \alpha x + \beta + \int_{\mathbb{R}} (x-u)_+ \, \mu(\mathrm{d}u), \tag{9}$$

where $\alpha, \beta \in \mathbb{R}$ and μ is a nonnegative Borel measure on $\mathbb{R}$ [SS07, Prop. 3.A.4]. Integrability of X, Y ensures that $\mathbb{E}[\alpha X + \beta] = \mathbb{E}[\alpha Y + \beta]$. Tonelli's Theorem and (8) yield that

$$\mathbb{E}[f(X)] = \alpha\mathbb{E}[X] + \beta + \int_{\mathbb{R}} \mathbb{E}[(X-u)_+] \, \mu(\mathrm{d}u) \le \alpha\mathbb{E}[Y] + \beta + \int_{\mathbb{R}} \mathbb{E}[(Y-u)_+] \, \mu(\mathrm{d}u) = \mathbb{E}[f(Y)],$$

whenever $\mathbb{E}[f(Y)] < \infty$. This is the desired convex domination. □

We next collect three elementary lemmas that will be used repeatedly in subsequent sections.

Lemma 5 (Layer-cake for hinges)**.** *Let X be integrable and let $u \in \mathbb{R}$. Then*

$$\mathbb{E}[(X-u)_+] = \int_u^\infty \mathbb{P}(X > t) \, \mathrm{d}t.$$

Proof. This is the layer-cake identity applied to the nonnegative random variable $(X-u)_+$, namely

$$\mathbb{E}[(X-u)_+] = \int_0^\infty \mathbb{P}((X-u)_+ > s) \, \mathrm{d}s = \int_0^\infty \mathbb{P}(X > u+s) \, \mathrm{d}s = \int_u^\infty \mathbb{P}(X > t) \, \mathrm{d}t.$$

□

Lemma 6 (Tangent line lower bound)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $g : I \to \mathbb{R}$ be convex and differentiable. Fix $B \in \mathbb{R}$ and $p_0 \in \mathbb{R}$. If there exists $u_\star \in I$ such that $g(u_\star) = B - p_0 u_\star$ and $g'(u_\star) = -p_0$, then*

$$g(u) \ge B - p_0 u \qquad \text{for all } u \in I.$$

Proof. By convexity, for all $u \in I$,

$$g(u) \ge g(u_\star) + g'(u_\star)(u - u_\star) = B - p_0 u.$$

□

The following monotone-ratio principle will allow us to deduce $D \ge 0$ from the sign pattern of D'.

Lemma 7 (Monotone-ratio principle)**.** *Let $D : [a, \infty) \to \mathbb{R}$ be differentiable with $\lim_{u\to\infty} D(u) = 0$. Assume $D'(u) = w(u)\big(1 - R(u)\big)$ for $u \ge a$, where $w(u) > 0$ and R is nondecreasing. If $D(a) \ge 0$, then $D(u) \ge 0$ for all $u \ge a$.*

Proof. If D attains a negative value, let u_0 be a point where D achieves its minimum on $[a, \infty)$. Then $D(u_0) < 0$ and necessarily $u_0 > a$ and $D'(u_0) = 0$. Since $w(u_0) > 0$, we have $R(u_0) = 1$. By monotonicity of R, $R(u) \le 1$ for $u \le u_0$ and $R(u) \ge 1$ for $u \ge u_0$. As such, $D'(u) \ge 0$ for $u \le u_0$ and $D'(u) \le 0$ for $u \ge u_0$, so that u_0 is instead a global maximum of D on $[a, \infty)$, contradicting the assumption that $D(u_0) < 0$ and $\lim_{u\to\infty} D(u) = 0$. □

3 A sharp stop-loss envelope under the two-sided tail constraint

Fix a deterministic function $s : [0, \infty) \to [0, 1]$. We interpret s as a two-sided tail envelope: an integrable random variable X 'satisfies the tail constraint s' if

$$\mathbb{P}(|X| > t) \le s(t) \qquad \text{for all } t \ge 0.$$

Lemma 8 gives a sharp upper envelope J_s for the stop-loss transform $u \mapsto \mathbb{E}[(X - u)_+]$ over all mean-zero X obeying this constraint. The sub-Gaussian and sub-exponential comparisons later are obtained by specializing s to $s_G : t \mapsto \min\{1, 2e^{-t^2/2}\}$ and $s_E : t \mapsto \min\{1, 2e^{-t}\}$, respectively.

Lemma 8 (Sharp stop-loss envelope)**.** *Let $s : [0, \infty) \to [0, 1]$ be non-increasing and continuous, and assume*

$$A_s := \int_0^\infty s(t)\,\mathrm{d}t < \infty, \qquad B_s := \frac{A_s}{2}.$$

Define

$$H_s(x) := x\,s(x) + \int_x^\infty s(t)\,\mathrm{d}t.$$

Assume there exists $a > 0$ such that $H_s(a) = B_s$, and set $p_0 := s(a)$. Define $J_s : [0, \infty) \to \mathbb{R}_+$ by

$$J_s(u) := \begin{cases} B_s - p_0 u, & 0 \le u \le a, \\ \int_u^\infty s(t)\,\mathrm{d}t, & u \ge a. \end{cases}$$

If X is integrable with $\mathbb{E}[X] = 0$ and satisfies the tail constraint

$$\mathbb{P}\big(|X| > t\big) \le s(t) \qquad \textit{for all } t \ge 0,$$

then for every $u \ge 0$, there holds the stop-loss bound

$$\mathbb{E}[(X - u)_+] \le J_s(u).$$

Proof. Let $p(t) := \mathbb{P}(X > t)$. Then, by assumption, p is nonincreasing and $p(t) \le s(t)$ for all $t \ge 0$. By Lemma 5, we compute

$$\int_0^\infty p(t)\,\mathrm{d}t = \mathbb{E}[X_+].$$

Since $\mathbb{E}[X] = 0$, it holds that $\mathbb{E}|X| = 2\,\mathbb{E}[X_+]$. Another application of Lemma 5 yields the bound

$$\mathbb{E}|X| = \int_0^\infty \mathbb{P}(|X| > t)\,\mathrm{d}t \le \int_0^\infty s(t)\,\mathrm{d}t = A_s,$$

and hence $\int_0^\infty p(t)\,\mathrm{d}t \le A_s/2 = B_s$. Fix now $u \ge 0$ and set $\alpha := p(u)$. Since p is nonincreasing, we can make the elementary bound

$$\int_0^u p(t)\,\mathrm{d}t \ge \alpha u, \qquad \text{hence} \qquad \int_u^\infty p(t)\,\mathrm{d}t \le B_s - \alpha u. \tag{M}$$

Separately, since $p(t) \le \alpha$ for $t \ge u$, we can equally bound

$$\int_u^\infty p(t)\,\mathrm{d}t \le \int_u^\infty \min\{\alpha, s(t)\}\,\mathrm{d}t. \tag{C}$$

If $u \geq a$, then (C) gives immediately that $\int_u^\infty p(t)\,\mathrm{d}t \leq \int_u^\infty s(t)\,\mathrm{d}t = J_s(u)$.

Assume then that $u < a$. If $\alpha \geq p_0$, then (M) yields that

$$\int_u^\infty p(t)\,\mathrm{d}t \leq B_s - \alpha u \leq B_s - p_0 u = J_s(u).$$

Finally, assume that $u < a$ and $\alpha < p_0$. If $\alpha = 0$, then $p(t) = 0$ for all $t \geq u$, so $\int_u^\infty p(t)\,\mathrm{d}t = 0 \leq J_s(u)$ and there is nothing to prove. Assume therefore that $\alpha > 0$. By continuity and monotonicity of s and $\lim_{t\to\infty} s(t) = 0$, there exists $b > a$ such that $s(b) = \alpha$. (C) then gives that

$$\int_u^\infty p(t)\,\mathrm{d}t \leq \alpha(b-u) + \int_b^\infty s(t)\,\mathrm{d}t = H_s(b) - \alpha u.$$

Since s is nonincreasing and $s(b) = \alpha$, we have the elementary bound $\int_a^b s(t)\,\mathrm{d}t \geq \alpha(b-a)$, and can hence deduce that

$$\begin{aligned} B_s - H_s(b) = H_s(a) - H_s(b) &= ap_0 - b\alpha + \int_a^b s(t)\,\mathrm{d}t \\ &\geq ap_0 - b\alpha + \alpha(b-a) = a(p_0 - \alpha) \geq u(p_0 - \alpha), \end{aligned}$$

using $u < a$ in the last step. Rearrangement then gives that $H_s(b) - \alpha u \leq B_s - p_0 u = J_s(u)$, and we conclude. □

Lemma 9 (A global linear lower bound for J_s)**.** *Work in the setting of Lemma 8. Then*

$$J_s(u) \geq B_s - p_0 u \qquad \textit{for all } u \geq 0.$$

Proof. By definition, $J_s(u) = B_s - p_0 u$ for $u \in [0, a]$. For $u \geq a$, we can write $J_s(u) = \int_u^\infty s(t)\,\mathrm{d}t$ and $J_s(a) = B_s - p_0 a$. Since s is nonincreasing, we can decompose

$$J_s(u) = J_s(a) - \int_a^u s(t)\,\mathrm{d}t \geq B_s - p_0 a - p_0(u-a) = B_s - p_0 u.$$

□

Lemma 10 (Extremizer attaining J_s)**.** *Work in the setting of Lemma 8, and assume in addition that there exists $t_0 \geq 0$ such that $s(t) = 1$ for $t \in [0, t_0]$ and s is strictly decreasing on $[t_0, \infty)$. Assume also that the solution a to $H_s(a) = B_s$ satisfies $a > t_0$. Let $X^\star$ be the random variable with distribution function*

$$F_\star(x) := \begin{cases} 0, & x \leq -a, \\ s(-x) - p_0, & -a < x \leq -t_0, \\ 1 - p_0, & -t_0 < x < a, \\ 1 - s(x), & x \geq a. \end{cases}$$

The function $F_\star$ is a cumulative distribution function on $\mathbb{R}$. Then $\mathbb{P}(|X^\star| > t) = s(t)$ for all $t \geq 0$ and $\mathbb{E}[X^\star] = 0$. Moreover, for every $u \geq 0$, the stop-loss satisfies

$$\mathbb{E}[(X^\star - u)_+] = J_s(u).$$

Proof. We first verify the two-sided tail by considering cases. If $0 \le t < t_0$, then $X^\star \notin (-t_0, a)$ almost surely, so $\mathbb{P}(|X^\star| > t) = 1 = s(t)$. If $t_0 \le t < a$, then $\mathbb{P}(X^\star > t) = p_0$ and

$$\mathbb{P}(X^\star < -t) = F_\star(-t) = s(t) - p_0,$$

and so $\mathbb{P}(|X^\star| > t) = s(t)$. If $t \ge a$, then $\mathbb{P}(X^\star > t) = s(t)$ and $\mathbb{P}(X^\star < -t) = 0$, so again $\mathbb{P}(|X^\star| > t) = s(t)$.

By Lemma 5 and the identity $\mathbb{P}(|X^\star| > t) = s(t)$, compute that

$$\mathbb{E}|X^\star| = \int_0^\infty \mathbb{P}(|X^\star| > t)\,\mathrm{d}t = \int_0^\infty s(t)\,\mathrm{d}t = A_s,$$

and also that

$$\mathbb{E}[X_+^\star] = \int_0^\infty \mathbb{P}(X^\star > t)\,\mathrm{d}t = \int_0^a p_0\,\mathrm{d}t + \int_a^\infty s(t)\,\mathrm{d}t = ap_0 + \int_a^\infty s(t)\,\mathrm{d}t = H_s(a) = B_s.$$

We thus see that (writing $X_-^\star = \max(-X^\star, 0)$) $\mathbb{E}[X_-^\star] = \mathbb{E}|X^\star| - \mathbb{E}[X_+^\star] = A_s - B_s = B_s$, and hence that $\mathbb{E}[X^\star] = 0$.

Finally, for $u \ge 0$, Lemma 5 gives that

$$\mathbb{E}[(X^\star - u)_+] = \int_u^\infty \mathbb{P}(X^\star > t)\,\mathrm{d}t.$$

If $u < a$, then this equals $p_0(a-u) + \int_a^\infty s(t)\,\mathrm{d}t = B_s - p_0 u = J_s(u)$, whereas if $u \ge a$, then it equals $\int_u^\infty s(t)\,\mathrm{d}t = J_s(u)$, i.e. equality holds throughout. □

Proposition 11 (From envelope domination to convex domination)**.** *Work in the setting of Lemma 8 and write J_s for the corresponding envelope. Let X be integrable with $\mathbb{E}[X] = 0$ and satisfy the tail constraint $\mathbb{P}(|X| > t) \le s(t)$ for all $t \ge 0$. Let Y be integrable and symmetric about 0, and set $g_Y(u) := \mathbb{E}[(Y-u)_+]$. If*

$$g_Y(u) \ge J_s(u) \qquad \text{for all } u \ge 0,$$

then $X \preceq_{cx} Y$.

Proof. For $u \ge 0$, Lemma 8 gives $\mathbb{E}[(X-u)_+] \le J_s(u) \le g_Y(u) = \mathbb{E}[(Y-u)_+]$. Applying Lemma 8 to $-X$ (which also has $\mathbb{E}[-X] = 0$ and satisfies the same tail constraint as X) yields $\mathbb{E}[(-X-u)_+] \le J_s(u) \le g_Y(u) = \mathbb{E}[(-Y-u)_+]$ for all $u \ge 0$, using symmetry of Y. Fix $u \in \mathbb{R}$. If $u \ge 0$ then the preceding display gives $g_X(u) \le g_Y(u)$. If $u < 0$, then set $v := -u > 0$ and use the identity

$$(x+v)_+ = (-x-v)_+ + x + v$$

to write

$$g_X(-v) = \mathbb{E}[(X+v)_+] = \mathbb{E}[(-X-v)_+] + v = g_{-X}(v) + v, \qquad g_Y(-v) = g_{-Y}(v) + v,$$

since $\mathbb{E}[X] = \mathbb{E}[Y] = 0$. The bound for $-X$ yields $g_{-X}(v) \le g_{-Y}(v)$ for all $v \ge 0$, and hence $g_X(u) \le g_Y(u)$ for all $u \in \mathbb{R}$. Applying Proposition 4 then completes the proof. □

4 Gaussian domination and proof of Theorem 1

By Proposition 11, Theorem 1 follows once we show that $g_{c_0}(u) \geq J_G(u)$ for all $u \geq 0$, where J_G is the stop-loss envelope from Lemma 8 for the sub-Gaussian tail envelope $s_G : t \mapsto \min\{1, 2e^{-t^2/2}\}$. This section proves this inequality and identifies the sharp c_0.

For $c > 0$ define the Gaussian stop-loss transform

$$g_c(u) := \mathbb{E}[(cG - u)_+], \qquad u \in \mathbb{R}. \tag{10}$$

We first recall an exact formula for g_c.

Lemma 12 (Gaussian stop-loss formula)**.** *For $c > 0$ and $u \in \mathbb{R}$,*

$$g_c(u) = c\,\varphi(u/c) - u\,\overline{\Phi}(u/c). \tag{11}$$

In particular, g_c is convex, differentiable, and satisfies

$$g_c'(u) = -\overline{\Phi}(u/c). \tag{12}$$

Proof. Let $Z := cG$. By Lemma 5,

$$g_c(u) = \int_u^\infty \mathbb{P}(Z > t)\,\mathrm{d}t = \int_u^\infty \overline{\Phi}(t/c)\,\mathrm{d}t.$$

Differentiate to obtain $g_c'(u) = -\overline{\Phi}(u/c)$. Integrating by parts in the last display gives (11). □

Remark 13 (Crude bounds)**.** *We record a short verification of $a > \sqrt{2}$ and $c_0 > \sqrt{2}$; these numerical bounds will be used in subsequent developments. First, because $H(a) = B$ and H is strictly monotone, it suffices to show that $H(\sqrt{2}) > B$. Since $H(\sqrt{2}) > 2\sqrt{2}e^{-1}$ and*

$$B = \frac{t_0}{2} + \int_{t_0}^\infty e^{-t^2/2}\,\mathrm{d}t \leq \frac{t_0}{2} + \frac{e^{-t_0^2/2}}{t_0} = \frac{t_0}{2} + \frac{1}{2t_0}, \qquad t_0 = \sqrt{2\log 2} \in (1.17, 1.18),$$

one checks readily that $B < 1.02 < 2\sqrt{2}e^{-1} < H(\sqrt{2})$, and hence that $a > \sqrt{2}$. Similarly, setting $x_1 := \sqrt{2\log 4}$ so that $e^{-x_1^2/2} = 1/4$, one writes $H(x_1) = \sqrt{\log 2} + 2\int_{x_1}^\infty e^{-t^2/2}\,\mathrm{d}t$. The bound $\varphi(x)/\overline{\Phi}(x) \leq x + 1/x$ from [AS64, Eq. 7.1.13] rearranges to $\int_x^\infty e^{-t^2/2}\,\mathrm{d}t \geq \frac{x}{x^2+1}\,e^{-x^2/2}$, whence $H(x_1) \geq \sqrt{\log 2} \cdot \frac{4\log 2+2}{4\log 2+1} > 1.05 > B$, and so $a > x_1$. Consequently $p_0 = 2e^{-a^2/2} < 2e^{-\log 4} = 1/2$, confirming $z > 0$. For the second inequality, use that $\varphi(z) \leq \varphi(0) = (2\pi)^{-1/2}$ and $B \geq t_0/2 > 0.58$ to see that $c_0 = B/\varphi(z) \geq B\sqrt{2\pi} > 1.45 > \sqrt{2}$.

Lemma 14 (A monotone ratio)**.** *Define*

$$R(u) := \frac{\overline{\Phi}(u/c_0)}{2e^{-u^2/2}} \qquad (u \geq a).$$

Then R is nondecreasing on $[a, \infty)$.

Proof. Differentiate $\log R(u) = \log \overline{\Phi}(u/c_0) + u^2/2 - \log 2$ to obtain that

$$\frac{d}{du}\log R(u) = u - \frac{1}{c_0}\frac{\varphi(u/c_0)}{\overline{\Phi}(u/c_0)}.$$

The Mills ratio bound $\varphi(x)/\overline{\Phi}(x) \le x + 1/x$ for $x > 0$ [AS64, Eq. 7.1.13] yields that for $u \ge a$,

$$\frac{d}{du} \log R(u) \ge u - \frac{1}{c_0}\left(\frac{u}{c_0} + \frac{c_0}{u}\right) = u\Big(1 - \frac{1}{c_0^2}\Big) - \frac{1}{u}.$$

By Remark 13, one sees that for $u \ge a$, it holds that $u^2(1 - 1/c_0^2) > 1$, and so this expression is strictly positive. It thus follows that R is increasing on this same interval. □

Lemma 15 (Gaussian stop-loss dominates the envelope)**.** *For all $u \ge 0$, we have $g_{c_0}(u) \ge J_G(u)$.*

Proof. Set $u_\star := c_0 z$. By Lemma 12, g_{c_0} is convex and differentiable on $\mathbb{R}$ with $g'_{c_0}(u) = -\overline{\Phi}(u/c_0)$. Since $\overline{\Phi}(z) = p_0$, we check that $g'_{c_0}(u_\star) = -p_0$. Moreover,

$$g_{c_0}(u_\star) = c_0\,\varphi(z) - u_\star\,\overline{\Phi}(z) = c_0\,\varphi(z) - p_0 u_\star = B - p_0 u_\star,$$

using the definition $c_0 = B/\varphi(z)$. Lemma 6 therefore yields that

$$g_{c_0}(u) \ge B - p_0 u \qquad \text{for all } u \in \mathbb{R}. \tag{13}$$

In particular, $g_{c_0}(u) \ge J_G(u)$ for $u \in [0, a]$.

For $u \ge a$, define then the difference

$$D(u) := g_{c_0}(u) - \int_u^\infty s_G(t)\,\mathrm{d}t.$$

By Lemma 12, D is differentiable and (noting that $u \ge a > t_0$, whereby $s_G(t) = 2e^{-t^2/2}$) we can compute

$$D'(u) = -\overline{\Phi}(u/c_0) + 2e^{-u^2/2} = 2e^{-u^2/2}\big(1 - R(u)\big),$$

where $R(u) := \overline{\Phi}(u/c_0)\big/(2e^{-u^2/2})$. By Lemma 14, the function R is increasing on $[a, \infty)$. Moreover, by (13) and the identity $\int_a^\infty s_G(t)\,\mathrm{d}t = B - ap_0$ (using that $H(a) = B$), we can check that

$$D(a) = g_{c_0}(a) - \int_a^\infty s_G(t)\,\mathrm{d}t \ge (B - ap_0) - (B - ap_0) = 0.$$

By inspection, one checks that $\lim_{u\to\infty} D(u) = 0$, and Lemma 7 therefore implies that $D(u) \ge 0$ for all $u \ge a$, i.e. $g_{c_0}(u) \ge J_G(u)$ as claimed. □

Proof of Theorem 1. Let $Y := c_0 G$. Lemma 15 gives that $g_{c_0}(u) = \mathbb{E}[(Y - u)_+] \ge J_G(u)$ for all $u \ge 0$. Proposition 11 therefore yields $X \preceq_{cx} Y$.

For sharpness, let $X^\star$ be the extremizer from Lemma 10 for the sub-Gaussian envelope $s_G(t) = \min\{1, 2e^{-t^2/2}\}$, so that $\mathbb{P}(|X^\star| > t) = s_G(t)$ for all $t \ge 0$ and $\mathbb{E}[(X^\star - u)_+] = J_G(u)$ for all $u \ge 0$. Fix $c \in (0, c_0)$ and set $u_c := cz$, where $z = \Phi^{-1}(1 - p_0)$ as before. Lemma 9 yields that $J_G(u_c) \ge B - p_0 u_c$. Using Lemma 12 and that $\overline{\Phi}(z) = p_0$, compute that

$$\mathbb{E}[(X^\star - u_c)_+] - \mathbb{E}[(cG - u_c)_+] = J_G(u_c) - g_c(u_c) \ge (B - p_0 u_c) - \big(c\,\varphi(z) - p_0 u_c\big) = B - c\,\varphi(z) > 0,$$

because $c < c_0 = B/\varphi(z)$. Taking $f(x) = (x - u_c)_+$ thus demonstrates that $X^\star \not\preceq_{cx} cG$, and it hence follows that the constant is unimprovable, i.e. $c_\star = c_0$. □

5 Laplace domination under sub-exponential tail constraints

Using the same tools, an analogous comparison is viable for random variables adhering to other tail constraints. In this section, we develop such a result under the assumption of a two-sided sub-exponential tail envelope

$$s_E(t) := \min\{1, 2e^{-t}\}, \qquad t \geq 0 \tag{14}$$

for which the natural comparator is a scaled standard Laplace random variable L with density $\frac{1}{2}e^{-|x|}$ on $\mathbb{R}$.

In particular, by Proposition 11, the sharp constant for the analog to Theorem 1 can be determined by identifying the minimal $c_E > 0$ for which $g_{c_E L}(u) \geq J_E(u)$ for all $u \geq 0$, with g the exact stop-loss for the Laplace random variable, and J_E the stop-loss envelope for the family of random variables satisfying the sub-exponential tail constraint.

Towards establishing such a comparison, define

$$A_E := \int_0^\infty s_E(t)\,\mathrm{d}t, \qquad B_E := \frac{A_E}{2}, \qquad H_E(x) := x\,s_E(x) + \int_x^\infty s_E(t)\,\mathrm{d}t.$$

Let $a_E > 0$ satisfy $H_E(a_E) = B_E$ and set $p_E := s_E(a_E)$. Since $H_E(x) = 2e^{-x}(x+1)$ for $x > \log 2$ and $B_E = (\log 2 + 1)/2$, one checks that $H_E(2\log 2) = (2\log 2 + 1)/2 > B_E$, so $a_E > 2\log 2$ and $p_E = 2e^{-a_E} < 1$. Let J_E denote the envelope J_{s_E} from Lemma 8. Finally, define

$$w_E := \log\frac{1}{2p_E}, \qquad c_E := \frac{B_E}{p_E(1+w_E)}. \tag{15}$$

Note that $w_E = a_E - 2\log 2$ and $c_E > 1$. In particular, some extended (but elementary) calculations yield that $c_E \approx 1.89389433$.

Theorem 16 (Sharp one-dimensional convex sub-exponential comparison)**.** *Let X be integrable with $\mathbb{E}[X] = 0$ and assume the tail constraint*

$$\mathbb{P}(|X| > t) \leq \min\{1, 2e^{-t}\} \qquad \textit{for all } t \geq 0.$$

Then $X \preceq_{cx} c_E L$. Moreover, c_E is optimal: for every $c < c_E$ there exists an integrable mean-zero X satisfying the same tail bound but with $X \not\preceq_{cx} cL$.

Lemma 17 (Laplace stop-loss transform)**.** *For $c > 0$, define $\ell_c(u) := \mathbb{E}[(cL - u)_+]$. Then for all $u \geq 0$, one has the exact formulae*

$$\ell_c(u) = \frac{c}{2}e^{-u/c}, \qquad \ell_c'(u) = -\frac{1}{2}e^{-u/c} = -\mathbb{P}(cL > u).$$

In particular, ℓ_c is convex and decreasing on $[0, \infty)$.

Proof. For $u \geq 0$, compute that

$$\ell_c(u) = \int_{u/c}^\infty (cx - u)\,\frac{1}{2}e^{-x}\,\mathrm{d}x = c\int_{u/c}^\infty (x - u/c)\,\frac{1}{2}e^{-x}\,\mathrm{d}x = \frac{c}{2}e^{-u/c}.$$

Differentiating gives the expression for $\ell_c'(u)$. □

Lemma 18 (Laplace stop-loss dominates the envelope)**.** *For all $u \geq 0$, we have $\ell_{c_E}(u) \geq J_E(u)$.*

Proof. Set $u_\star := c_E w_E$. By Lemma 17, ℓ_{c_E} is convex and differentiable on $[0,\infty)$ with $\ell'_{c_E}(u) = -\frac{1}{2}e^{-u/c_E}$. Since $\mathbb{P}(L > w_E) = \frac{1}{2}e^{-w_E} = p_E$, we have $\ell'_{c_E}(u_\star) = -p_E$. Moreover, by the definition of c_E, we see that

$$\ell_{c_E}(u_\star) = \frac{c_E}{2}e^{-w_E} = c_E p_E = B_E - p_E u_\star.$$

Lemma 6 therefore yields that $\ell_{c_E}(u) \geq B_E - p_E u$ for all $u \geq 0$, and hence that $\ell_{c_E}(u) \geq J_E(u)$ for $u \in [0, a_E]$.

For $u \geq a_E$, define the difference

$$D(u) := \ell_{c_E}(u) - \int_u^\infty s_E(t)\,\mathrm{d}t.$$

Since $a_E > \log 2$, we have $s_E(u) = 2e^{-u}$ for all $u \geq a_E$. Lemma 17 then gives that

$$D'(u) = -\frac{1}{2}e^{-u/c_E} + 2e^{-u} = 2e^{-u}\big(1 - R(u)\big), \qquad R(u) := \frac{1}{4}\,e^{u(1-1/c_E)}.$$

In particular, since $c_E > 1$, one sees that R is increasing on $[a_E, \infty)$ (and even on all of $\mathbb{R}$). Also, since $H_E(a_E) = B_E$ and $s_E(a_E) = p_E$, we can check that

$$\int_{a_E}^\infty s_E(t)\,\mathrm{d}t = B_E - a_E p_E \qquad \text{and} \qquad J_E(a_E) = B_E - p_E a_E = B_E - a_E p_E.$$

We thus see that $D(a_E) = \ell_{c_E}(a_E) - J_E(a_E) \geq 0$ by the first part of the proof. Moreover, by inspection, $\lim_{u\to\infty} D(u) = 0$, and so Lemma 7 therefore implies that $D(u) \geq 0$ for all $u \geq a_E$, i.e. $\ell_{c_E}(u) \geq J_E(u)$ as claimed. □

Proof of Theorem 16. Let $Y := c_E L$. Lemma 18 gives that $g_Y(u) = \mathbb{E}[(Y-u)_+] = \ell_{c_E}(u) \geq J_E(u)$ for all $u \geq 0$. Proposition 11 therefore yields that $X \preceq_{cx} Y$.

For sharpness, let $X^\star$ be the extremizer from Lemma 10 applied to the envelope s_E. Then $\mathbb{P}(|X^\star| > t) = s_E(t)$ for all $t \geq 0$ and $\mathbb{E}[(X^\star - u)_+] = J_E(u)$ for all $u \geq 0$. Fix $c \in (0, c_E)$ and set $u_c := c w_E$. Lemma 9 then yields that $J_E(u_c) \geq B_E - p_E u_c$. Using Lemma 17 and $\frac{1}{2}e^{-w_E} = p_E$, compute that

$$\mathbb{E}[(X^\star - u_c)_+] - \mathbb{E}[(cL - u_c)_+] = J_E(u_c) - \ell_c(u_c) \geq (B_E - p_E u_c) - c\,p_E = B_E - c p_E(1 + w_E) > 0,$$

because $c < c_E = B_E/(p_E(1+w_E))$. Taking $f(x) = (x - u_c)_+$ then shows that $X^\star \not\preceq_{cx} cL$, as claimed. □

6 Higher-dimensional consequences

Theorem 1 is one-dimensional: our proof relies heavily on the stop-loss characterisation of the convex ordering, which does not immediately extend to higher dimension. Nevertheless, we record two applications to stylised high-dimensional settings. The first assumes a meaningful coordinate structure (a martingale decomposition); the second orders expectations only among a restricted class of convex functions.

Tensorization via a sequential (martingale) coordinate representation

Write $[d] := \{1, \dots, d\}$. Throughout this subsection, $(\mathcal{F}_i)_{i=0}^d$ is a filtration.

Theorem 19 (Sequential tensorization for convex domination)**.** *Let $d \geq 1$. Let $X_1, \dots, X_d$ be integrable real random variables such that X_i is $\mathcal{F}_i$-measurable. Let $Y_1, \dots, Y_d$ be integrable real random variables that are mutually independent and independent of $\mathcal{F}_d$. Assume that for each $i \in [d]$ and every convex $\phi : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{E}[\phi(X_i) \,|\, \mathcal{F}_{i-1}] \leq \mathbb{E}[\phi(Y_i)] \qquad a.s. \tag{16}$$

Then, for every convex $f : \mathbb{R}^d \to \mathbb{R}$, it holds that

$$\mathbb{E}\big[f(X_1, \dots, X_d)\big] \leq \mathbb{E}\big[f(Y_1, \dots, Y_d)\big], \tag{17}$$

where both sides are understood as extended expectations in $(-\infty, \infty]$.

Proof. Fix a convex $f : \mathbb{R}^d \to \mathbb{R}$. Since f is convex and proper, we are free to choose an affine minorant $\ell(x) = a^\top x + b \leq f(x)$ and set $h := f - \ell \geq 0$. Since ℓ is affine, invoking (16) with $\phi(t) = t$ and $\phi(t) = -t$ gives that $\mathbb{E}[X_i \mid \mathcal{F}_{i-1}] = \mathbb{E}[Y_i]$ a.s., and hence that $\mathbb{E}[\ell(X)] = \mathbb{E}[\ell(Y)]$. It suffices to prove (17) with f replaced by h, i.e. to restrict attention to non-negative convex test functions.

Define $h_d := h$ and for $i = 1, \dots, d$ set

$$h_{i-1}(x_1, \dots, x_{i-1}) := \mathbb{E}\big[h_i(x_1, \dots, x_{i-1}, Y_i)\big].$$

Each h_i is convex and nonnegative. Fix $i \in [d]$. Since $(X_1, \dots, X_{i-1})$ is $\mathcal{F}_{i-1}$-measurable and $t \mapsto h_i(X_1, \dots, X_{i-1}, t)$ is convex, (16) yields that

$$\mathbb{E}[h_i(X_1, \dots, X_{i-1}, X_i) \,|\, \mathcal{F}_{i-1}] \leq \mathbb{E}\big[h_i(X_1, \dots, X_{i-1}, Y_i)\big] = h_{i-1}(X_1, \dots, X_{i-1}),$$

using independence of Y_i from $\mathcal{F}_{i-1}$. Taking expectations then gives that $\mathbb{E}[h_i(X_1, \dots, X_i)] \leq \mathbb{E}[h_{i-1}(X_1, \dots, X_{i-1})]$. Iterating from $i = d$ down to $i = 1$ yields that $\mathbb{E}[h(X)] \leq \mathbb{E}[h_0]$. By construction and independence of the Y_i, $\mathbb{E}[h_0] = \mathbb{E}[h(Y)]$, and so we conclude. □

Lemma 20 (Conditional form of Theorem 1)**.** *Let $\mathcal{G}$ be a sub-σ-field and let Z be integrable. Assume that $\mathbb{E}[Z \mid \mathcal{G}] = 0$ a.s. and*

$$\mathbb{P}\big(|Z| > t \mid \mathcal{G}\big) \leq 2e^{-t^2/2} \qquad a.s. \text{ for all } t \geq 0.$$

Let $G \sim \mathcal{N}(0, 1)$ be independent of $\mathcal{G}$. Then for every convex $\phi : \mathbb{R} \to \mathbb{R}$, it holds that

$$\mathbb{E}[\phi(Z) \,|\, \mathcal{G}] \leq \mathbb{E}\big[\phi(c_0 G)\big] \qquad a.s.$$

Proof. Fix convex ϕ and choose an affine minorant $\ell(t) = \alpha t + \beta \leq \phi(t)$. Set $\psi := \phi - \ell \geq 0$. Since G is centered and $\mathbb{E}[Z \mid \mathcal{G}] = 0$, we have that $\mathbb{E}[\ell(Z) \mid \mathcal{G}] = \beta$ a.s. and $\mathbb{E}[\ell(c_0 G)] = \beta$. Now, let ν_ω be a regular conditional law of Z given $\mathcal{G}$. For $\mathbb{P}$-a.e. ω, the one-dimensional law ν_ω satisfies the tail constraint (1), so Theorem 1 applied to ν_ω yields that $\int \psi \, \mathrm{d}\nu_\omega \leq \mathbb{E}[\psi(c_0 G)]$. Since $\psi \geq 0$, $\int \psi \, \mathrm{d}\nu_\omega = \mathbb{E}[\psi(Z) \mid \mathcal{G}](\omega)$. Adding back ℓ gives the claim. □

Corollary 21 (Dimension-free domination from a martingale-coordinate representation)**.** *Fix an orthonormal basis $u_1, \dots, u_d$ of $\mathbb{R}^d$ and let $X \in \mathbb{R}^d$ be integrable. Define $\xi_i := \langle u_i, X \rangle$ and $\mathcal{F}_i := \sigma(\xi_1, \dots, \xi_i)$. Assume that for each $i \in [d]$, there hold the conditional centering and tail constraints*

$$\mathbb{E}[\xi_i \mid \mathcal{F}_{i-1}] = 0 \quad \text{and} \quad \mathbb{P}\big(|\xi_i| > t \mid \mathcal{F}_{i-1}\big) \leq \min\{1, 2e^{-t^2/2}\} \quad a.s. \text{ for all } t \geq 0.$$

Let $G \sim \mathcal{N}(0, I_d)$ be standard Gaussian in $\mathbb{R}^d$. Then for every convex $f : \mathbb{R}^d \to \mathbb{R}$, there holds the ordering

$$\mathbb{E}[f(X)] \leq \mathbb{E}[f(c_0 G)],$$

where both sides are understood as extended expectations in $(-\infty, \infty]$.

Proof. Let $(G_1, \ldots, G_d)$ be i.i.d. $\mathcal{N}(0,1)$ independent of $\mathcal{F}_d$ and set $Y_i := c_0 G_i$. Lemma 20 applied with $\mathcal{G} = \mathcal{F}_{i-1}$ and $Z = \xi_i$ yields that

$$\mathbb{E}[\phi(\xi_i) \,|\, \mathcal{F}_{i-1}] \leq \mathbb{E}[\phi(Y_i)] \qquad \text{a.s. for every convex } \phi : \mathbb{R} \to \mathbb{R}.$$

Applying then Theorem 19 to (overloading notation momentarily) $(X_1, \ldots, X_d) = (\xi_1, \ldots, \xi_d)$ and $(Y_1, \ldots, Y_d)$, we obtain that for every convex $\widetilde{f} : \mathbb{R}^d \to \mathbb{R}$,

$$\mathbb{E}[\widetilde{f}(\xi_1, \ldots, \xi_d)] \leq \mathbb{E}[\widetilde{f}(Y_1, \ldots, Y_d)].$$

Take $\widetilde{f}(z_1, \ldots, z_d) := f(\sum_{i=1}^d z_i u_i)$ to obtain that $\mathbb{E}[f(X)] \leq \mathbb{E}[f(\sum_{i=1}^d Y_i u_i)]$. Since $\sum_{i=1}^d G_i u_i \overset{d}{=} G$, the right-hand side equals $\mathbb{E}[f(c_0 G)]$. □

We emphasise that this is a genuine multivariate convex ordering, meaning that the conclusion holds for all convex $f : \mathbb{R}^d \to \mathbb{R}$.

Domination for the cone generated by convex ridge functions

Theorem 22 (Convex domination for nonnegative ridge combinations)**.** *Let $X \in \mathbb{R}^d$ be integrable and satisfy the vector tail bound*

$$\mathbb{E}[X] = 0 \qquad \text{and} \qquad \mathbb{P}\big(|\langle v, X\rangle| > t\big) \leq 2e^{-t^2/2} \quad \text{for all } v \in \mathbb{R}^d \text{ with } \|v\|_2 = 1,\ t \geq 0. \tag{18}$$

Let $G \sim \mathcal{N}(0, I_d)$. Fix a measurable $f : \mathbb{R}^d \to \mathbb{R}$ that admits a representation

$$f(x) = b + a^\top x + \sum_{k=1}^m \lambda_k \, \phi_k(\langle u_k, x\rangle), \tag{19}$$

with $m \in \mathbb{N}$, $a, u_k \in \mathbb{R}^d$, $b \in \mathbb{R}$, $\lambda_k \geq 0$, and each $\phi_k : \mathbb{R} \to \mathbb{R}$ convex, or is a pointwise increasing limit of such functions. Then there holds the ordering

$$\mathbb{E}[f(X)] \leq \mathbb{E}[f(c_0 G)],$$

where both sides are understood as extended expectations in $(-\infty, \infty]$.

Remark 23 (Ridge Convexity)**.** *The cone of functions considered herein could be termed the class of 'ridge-convex' functions. In dimensions $d \geq 2$, these form a strict subset of the cone of all convex functions on $\mathbb{R}^d$. As such, the 'linear convex' ordering established by Theorem 22 is in general strictly weaker than the 'full' convex ordering which one might seek (and indeed, which the result of [vH25] encourages one to seek).*

Proof. Step 1: the finite ridge case. Assume f has the form (19). For each k, pick an affine minorant $\ell_k(t) = \alpha_k t + \beta_k \leq \phi_k(t)$ and set $\psi_k := \phi_k - \ell_k \geq 0$. Rewrite

$$f(x) = b' + (a')^\top x + \sum_{k=1}^m \lambda_k \, \psi_k(\langle u_k, x\rangle),$$

where $b' := b + \sum_{k=1}^m \lambda_k \beta_k$ and $a' := a + \sum_{k=1}^m \lambda_k \alpha_k u_k$, i.e. we again reduce to the setting of non-negative convex test functions. Since $\psi_k \geq 0$ and $\lambda_k \geq 0$, Tonelli's Theorem gives that

$$\mathbb{E}[f(Z)] = b' + (a')^\top \mathbb{E}[Z] + \sum_{k=1}^m \lambda_k \, \mathbb{E}\big[\psi_k(\langle u_k, Z\rangle)\big] \qquad \text{for } Z \in \{X, c_0 G\}. \tag{20}$$

The affine term vanishes for both $Z = X$ and $Z = c_0 G$ by (18).

Fix k with $u_k \neq 0$ and set $v_k := u_k/\|u_k\|_2$. Then $Z_k := \langle v_k, X\rangle$ satisfies (1) by (18). Apply Theorem 1 to Z_k with the convex test function $\theta \mapsto \psi_k(\|u_k\|_2\, \theta)$ to obtain that

$$\mathbb{E}\big[\psi_k(\langle u_k, X\rangle)\big] \leq \mathbb{E}\big[\psi_k(\langle u_k, c_0 G\rangle)\big].$$

Summing over k in (20) yields that $\mathbb{E}[f(X)] \leq \mathbb{E}[f(c_0 G)]$ for ridge sums.

Step 2: monotone limits. Let f_n be ridge sums with $f_n \uparrow f$ pointwise. Choose an affine minorant $\ell \leq f_1$ and set $g_n := f_n - \ell \uparrow g := f - \ell$, so $g_n, g \geq 0$. Step 1 gives that $\mathbb{E}[g_n(X)] \leq \mathbb{E}[g_n(c_0 G)]$, and monotone convergence hence yields that $\mathbb{E}[g(X)] \leq \mathbb{E}[g(c_0 G)]$; adding back ℓ (whose expectations match) then gives the final claim. □

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